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Secondary cooling system for control modules

Abstract

A cooling system includes a reactant material, a reactive material, a thermally managed control module, and a plunger. The reactive material is spaced apart from the reactant material. The thermally managed control module is configured to receive coolant from a coolant supply reservoir via a coolant supply channel. The plunger is movable between a retracted position and an extended position. When a temperature of the thermally managed control module is greater than a predetermined temperature, an actuator is configured to transition the plunger from the retracted position to the extended position and cause the reactive material to mix with the reactant material to create a second coolant via an endothermic reaction, and the second coolant enters the thermally managed control module via the coolant supply channel.

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Background/Summary

INTRODUCTION

(1) The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(2) The present disclosure relates to a cooling system, and more specifically, to a secondary cooling system for control modules.

(3) Vehicles include one or more control modules. Some control modules run an algorithm to perform one or more functions. Examples of control modules include steering control modules, brake control modules, camera control modules, infotainment control modules, power inverter control modules, auxiliary power control modules, direct current (DC) to DC converter control modules, and another control modules.

(4) The control module may include a thermistor. The thermistor is configured to measure a temperature of the control module. When the control module performs the functions, a temperature of the control module may rise. In other words, the control module may generate heat when performing functions. The control module is configured to detect when the temperature of the control module is greater than a predetermined temperature.

(5) A cooling system may maintain the temperature of the control module below the predetermined temperature. The cooling system uses a coolant. The coolant is supplied to the control module via one or more coolant channels.

SUMMARY

(6) An example cooling system includes a reactant material, a reactive material, a thermally managed control module, and a plunger. The reactive material is spaced apart from the reactant material. The thermally managed control module is configured to receive coolant from a coolant supply reservoir via a coolant supply channel. The plunger movable between a retracted position and an extended position. When a temperature of the thermally managed control module is greater than a predetermined temperature, an actuator is configured to transition the plunger from the retracted position to the extended position and cause the reactive material to mix with the reactant material to create a second coolant via an endothermic reaction, and the second coolant enters the thermally managed control module via the coolant supply channel.

(7) In one example, the example cooling system includes a plunger channel in fluid communication with the coolant supply channel and housing the plunger and the actuator.

(8) In one example, the actuator is a ram. When the temperature of the thermally managed control module is greater than the predetermined temperature, the ram drives movement of the plunger from the retracted position to the extended position.

(9) In one example, the example cooling system includes a first sealing disk disposed between the

reactant material and the reactive material, and a second sealing disk disposed between the reactive material and the coolant supply channel.
(10) In one example, the plunger is disposed within the plunger channel when in the retracted position, and the plunger extends transversely across the coolant supply channel and ruptures the first and second sealing disks when transitioning from the retracted position to the extended position.
(11) In one example, the plunger includes a channel and the channel carries the second coolant to the coolant supply channel when the plunger is in the extended position.
(12) An example cooling system includes a fluid tank, a solute reservoir, and a thermally managed control module. The fluid tank contains a fluid. The solute reservoir is disposed within the fluid tank and containing an endothermic solute. The thermally managed control module has a coolant channel extending through the thermally managed control module and the coolant channel is configured to receive a coolant. When a temperature of the thermally managed control module is greater than a predetermined temperature, an actuator is configured to release the endothermic solute from the solute reservoir into the fluid tank and cause the endothermic solute to mix with the fluid to create a second coolant via an endothermic reaction, and the second coolant enters the thermally managed control module via a coolant supply channel.
(13) In one example, the example cooling system includes a gas cylinder and a release valve. The gas cylinder is attached to the solute reservoir and filled with a pressurized gas. The release valve is disposed between the gas cylinder and the solute reservoir and movable between an open position and a closed position. When the temperature of the thermally managed control module is greater than the predetermined temperature, the actuator is configured to transition the release valve from the closed position to the open position and allow the pressurized gas to flow from the gas cylinder to the solute reservoir and release the endothermic solute from the solute reservoir into the fluid.
(14) In one example, the example cooling system includes a bypass trigger disposed between the gas cylinder and the solute reservoir, and an airbag control module configured to activate the bypass trigger based on an activated state of an airbag. When the bypass trigger is activated, the bypass trigger is configured to allow the pressurized gas to flow from the gas cylinder to the solute reservoir and release the endothermic solute from the solute reservoir into the fluid tank to create the second coolant via the endothermic reaction, and the second coolant enters the coolant channel of the thermally managed control module via the coolant supply channel.
(15) In one example, the example cooling system includes a pump disposed outside the thermally managed control module and configured to pump the coolant from an outlet of the coolant channel to an inlet of the coolant channel.
(16) In one example, the example cooling system includes a mechanical valve disposed between the pump and the inlet of the coolant channel and movable between an open position and a closed position. When a coolant temperature of the coolant is above a predetermined coolant temperature, the coolant transitions the mechanical valve from the closed position to the open position and causes the endothermic solute to release from the solute reservoir into the fluid tank to create the second coolant via the endothermic reaction, and the second coolant enters the coolant channel of the thermally managed control module via the coolant supply channel.
(17) In one example, the thermally managed control module includes a cold plate and the coolant channel extends through the cold plate.
(18) In one example, the example cooling system includes the thermally managed control module includes a second coolant channel extending through a second cold plate and the second coolant channel is configured to receive the second coolant from the fluid tank via the coolant supply channel.
(19) An example cooling system includes a fluid tank, a thermally managed control module, and a coolant channel. The fluid tank is at least partially filled with a fluid. The fluid has a fluid temperature. The thermally managed control module is disposed within the fluid tank. The coolant

channel extends through the fluid tank and is configured to receive a coolant to cool the fluid in the fluid tank. When the fluid temperature of the fluid is greater than a predetermined temperature, an actuator is configured to cool the fluid temperature to less than the predetermined temperature by at least one of evaporating the fluid and mixing the fluid with an endothermic solute.
(20) In one example, the example cooling system includes a solute reservoir disposed within the fluid tank and containing the endothermic solute. When the fluid temperature is greater than the predetermined temperature, the actuator is configured to release the endothermic solute from the endothermic solute into the fluid of the fluid tank using a release assembly.
(21) In one example, the release assembly includes an impeller disposed adjacent to the solute reservoir and rotation of the impeller pierces a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
(22) In one example, the release assembly includes a magnet disposed inside the solute reservoir and a solenoid disposed outside the solute reservoir. The solenoid attracts the magnet and pierces a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
(23) In one example, the release assembly includes a fuse disposed inside the solute reservoir, and current traveling through the fuse is configured to melt a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
(24) In one example, the fluid tank is at least partially filled with a gas.
(25) In one example, the example cooling system includes a pressure relief valve movable between an open position and a closed position. When the fluid temperature is greater than the predetermined temperature, the fluid in the fluid tank evaporates and the gas causes the pressure relief valve to transition from the closed position into the open position to release the gas from the fluid tank via a vent.
(26) Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:
(2) FIG. **1** is a functional diagram of an example cooling system;
(3) FIG. **2** is a schematic diagram of an example cooling system in a first position;
(4) FIG. **3** is a schematic diagram of the example cooling system in a second position;
(5) FIG. **4** is a schematic diagram of the example cooling system in a third position;
(6) FIG. **5** is a schematic diagram of an example cooling system;
(7) FIG. **6** is a schematic diagram of an example cooling system;
(8) FIG. **7** is a schematic diagram of an example cooling system;
(9) FIG. **8** is a schematic diagram of an example cooling system;
(10) FIG. **9** is a schematic diagram of an example cooling system;
(11) FIG. **10** is a schematic diagram of an example cooling system;
(12) FIG. **11** is a schematic diagram of an example cooling system;
(13) FIG. **12** is a schematic diagram of an example release assembly;
(14) FIG. **13** is a schematic diagram of an example release assembly;
(15) FIG. **14** is a schematic diagram of an example release assembly; and
(16) FIG. **15** is a flow chart of an example method for operating the cooling system.
(17) In the drawings, reference numbers may be reused to identify similar and/or identical

elements.

DETAILED DESCRIPTION

(18) A vehicle includes wheels, a steering wheel, brakes, and a brake pedal. The steering wheel is operated by a driver of the vehicle to adjust a direction of the wheels, and therefore guide a direction of travel of the vehicle. The brake pedal is operated by the driver of the vehicle to activate the brakes and reduce a speed of the vehicle.

(19) Control modules execute instructions to perform functions. One example of a control module is a steering control module. The steering control module is configured to control the direction of the wheels based on an angular position of the steering wheel. In another example, the control module may be a brake control module. The brake control module is configured to control the brakes of the vehicle based on a position of the brake pedal of the vehicle. Other examples of control modules include camera control modules, infotainment control modules, power inverter control modules, auxiliary power control modules, DC to DC converter control modules, and another control module.

(20) When a control module performs functions, a temperature of the control module may rise. A cooling system may maintain the temperature of the control module below a predetermined temperature. The cooling system uses a coolant to cool the control module.

(21) However, the cooling system may not supply enough coolant to the control module to maintain the temperature of the control module below the predetermined temperature under some circumstances. For example, there may be a leak in cooling channels that supply coolant from a coolant reservoir to the control module. In another example, the coolant reservoir may have a little amount of coolant. In yet another example, a coolant pump configured to pump coolant from the coolant reservoir to the control module may fail and therefore, coolant may not be pumped for cooling.

(22) In view of the above, a secondary cooling system and a method for secondary cooling is provided for supplying coolant to the control module when the temperature of the control module is greater than the predetermined temperature.

(23) Referring now to FIG. **1**, a functional block diagram of an example cooling system **100** is presented. The example cooling system **100** includes a thermally managed control module **102**, a first cooling system **104**, and a second cooling system **106**.

(24) The cooling system **100** is described as a cooling system for a thermally managed control module of a vehicle. However, the cooling system may also be used for thermally managed control modules of non-vehicles.

(25) The first cooling system **104** includes a coolant reservoir, first coolant channels, and a coolant. The first coolant channels are in fluid communication with the coolant reservoir and the thermally managed control module **102**. Coolant is pumped from the coolant reservoir to the thermally managed control module **102** to cool the thermally managed control module **102**.

(26) In some configurations, the first cooling system **104** may include heat exchangers. Examples of heat exchanges include radiators, chillers or another suitable heat exchanger. The heat exchangers are configured to remove heat from the coolant and thereby cool the thermally managed control module **102**.

(27) When a temperature of the thermally managed control module **102** is greater than a predetermined temperature, an actuator **108** of the second cooling system **106** is configured to cause the temperature of the thermally managed control module **102** to decrease below the predetermined temperature. The actuator **108** may be a mechanical actuator, an electrical actuator, or a combination thereof. In one example, the actuator **108** of the second cooling system **106** is configured to cool the thermally managed control module **102** by triggering a reaction between a first material (e.g., solute, reactive material) and a second material (e.g., fluid, reactant material) to create an endothermic coolant, and the second cooling system **106** supplies the endothermic coolant to the thermally managed control module **102**. In another example, the actuator **108** of the second

cooling system **106** is configured to cause cooling of the thermally managed control module **102** by cooling (e.g., evaporating) a fluid in which the thermally managed control module **102** is disposed within.
(28) FIG. **2** is a schematic diagram of an example cooling system **120** in a first position. FIG. **3** is a schematic diagram of the example cooling system **120** in a second position. FIG. **4** is a schematic diagram of the example cooling system **120** in a third position.
(29) The cooling system **120** includes a thermally managed control module **122**, a first cooling system **124**, and a second cooling system **126**. The first cooling system **124** may include a coolant supply reservoir **130**, a coolant supply channel **132**, a coolant release channel **134**, and a coolant release reservoir **136**. The coolant supply channel **132** and the coolant release channel **134** may collectively be referred to as a coolant channel.
(30) The coolant supply channel **132** extends from the coolant supply reservoir **130** to the thermally managed control module **122**. The coolant supply channel **132** is in fluid communication with the thermally managed control module **122**. The coolant supply channel **132** carries coolant from the coolant supply reservoir **130** to the thermally managed control module **122**. In the illustrated example, the coolant supply channel **132** is shaped in a linear shape and positioned perpendicular to the thermally managed control module **122**. However, the coolant supply channel **132** may be shaped in another suitable shape and positioned in another suitable position.
(31) The coolant release channel **134** is in fluid communication with the thermally managed control module **122**. The coolant release channel **134** is attached to a coolant release reservoir **136**. In some configurations, the coolant release reservoir **136** is arranged in fluid communication with the coolant supply reservoir **130**.
(32) Using the first cooling system **124**, coolant is pumped from the coolant supply reservoir **130** to the thermally managed control module **122** via the coolant supply channel **132**. Coolant flows through the thermally managed control module **122** to cool the thermally managed control module **122**. More specifically, the coolant is used to maintain a temperature of the thermally managed control module **122** to below a predetermined temperature. Some coolant may be released to the coolant release reservoir **136** via the coolant release channel **134**.
(33) The second cooling system **126** includes a reservoir **140** containing a reactant material **142**, a reaction channel **144** containing a reactive material **146**, and a plunger channel **148** housing a plunger **150**.
(34) The reservoir **140** extends between a first reservoir end **154** and a second reservoir end **156** that is opposite the first reservoir end **154**. The reservoir **140** may be formed in a cylindrical shape or another suitable shape. The first reservoir end **154** is closed and the second reservoir end **156** is open to the reaction channel **144**. In other words, the reservoir **140** is in fluid communication with the reaction channel **144** via the second reservoir end **156**.
(35) A spring **158** is disposed within the reservoir **140** and positioned at the first reservoir end **154**. The spring **158** is movable along the reservoir **140**. The spring **158** includes a disk **160** and a spring coil **162**. The disk **160** sealingly engages with an inner surface **166** of the reservoir **140**. The spring coil **162** extends between the disk **160** and the first reservoir end **154** of the reservoir **140**. The spring coil **162** expands to drive movement of the disk **160**. More specifically, the disk **160** moves toward the second reservoir end **156** when the spring coil **162** of the spring **158** expands.
(36) The reactant material **142** is disposed within the reservoir **140** and the reaction channel **144**. Inside the reservoir **140**, the reactant material is disposed between the disk **160** of the spring **158** and the second reservoir end **156** of the reservoir **140**. The reactant material **142** may be water or another suitable reactant material. Movement of the spring **158** drives movement of the reactant material **142**. More specifically, expansion of the spring coil **162** causes the reactant material **142** to move from the reservoir **140** and reaction channel **144** toward the coolant supply channel **132**.
(37) The reaction channel **144** extends between a first channel end **170** and a second channel end **172** that is opposite to the first channel end **170**. The first channel end **170** is positioned adjacent to

the second reservoir end **156** of the reservoir **140**. The second channel end **172** is positioned adjacent to the coolant supply channel **132**.

(38) The reactive material **146** is positioned at the second channel end **172** of the reaction channel **144**. The reactive material **146** may be ammonium nitrite, urea, or another suitable reactive material. The reactive material **146** is sealed from the reactant material **142** using a first sealing disk **176** and sealed from the coolant supply channel **132** using a second sealing disk **178**.

(39) The plunger **150** is attached to a ram **182** and movable between a retracted position, as shown in FIG. **2**, and an extended position, as shown in FIGS. **3-4**. In the retracted position, the plunger **150** is positioned in the plunger channel **148**. In the extended position, the plunger **150** extends transversely across the coolant supply channel **132** and is positioned at least partially within the reaction channel **144**. When in the extended position, the plunger **150** blocks flow of coolant, if any, traveling in the coolant supply channel **132** from the coolant supply reservoir **130**. The plunger **150** ruptures the first and second sealing disks **176**, **178** when in the extended position.

(40) The plunger **150** extends between a first plunger end **184** and a second plunger end **186** that is opposite the first plunger end **184**. The plunger **150** includes a first plunger side **188** and a second plunger side **190** that is opposite the first plunger side **188**. A plunger nose **192** is disposed at the first plunger end **184**. The plunger **150** is attached to a ram **182** at the second plunger end **186**. In the illustrated example, the plunger **150** is sized and shaped to contour the plunger channel **148**. However, the plunger **150** may be another suitable size and another suitable shape. The ram **182** is one example of an actuator. The ram **182** drives movement of the plunger **150** from the retracted position to the extended position.

(41) A channel **196** extends through the plunger **150**. More specifically, the channel **196** extends between the first plunger end **184** and the second plunger side **190**. In the illustrated example, the channel **196** forms a right angle between the first plunger end **184** and the second plunger side **190**. However, the channel **196** may have another suitable shape, such as curved between the first plunger end **184** and the second plunger side **190**. When the plunger **150** is in the extended position, the first plunger end **184** of the plunger **150** is in fluid communication with the reaction channel **144** via the channel **196** and the second plunger side **190** of the plunger **150** is in fluid communication with the coolant supply channel **132** via the channel **196**.

(42) When a temperature of the thermally managed control module **122** is below a predetermined temperature as shown in FIG. **2**, the ram **182** is in an unactivated state and the plunger **150** is in the retracted position.

(43) When the temperature of the thermally managed control module **122** is greater than the predetermined temperature, the ram **182** transitions the plunger **150** from the retracted position to the extended position. In the extended position shown in FIG. **3**, the plunger **150** extends transversely across the coolant supply channel **132** and blocks flow, if any, of coolant flowing from the coolant supply reservoir **130** to the thermally managed control module **122** via the first cooling system **124**. The plunger nose **192** of the plunger **150** ruptures the first and second sealing disks **176**, **178**. As a result, the reactant and reactive materials **142**, **146** mix and create an endothermic coolant **200** via an endothermic reaction between the reactant material and the reactive material.

(44) As shown in FIG. **4**, the endothermic coolant **200** flows from the reaction channel **144** and into the channel **196** of the plunger **150**. As the endothermic coolant **200** begins to flow into the channel **196** of the plunger **150**, a pressure inside the reaction channel **144** lowers and the spring **158** expands to push the remaining reactant material **142** into the reactional channel **144** and the endothermic coolant **200** from inside the reservoir **140** toward the channel **196** of the plunger **150**. From the channel **196** of the plunger **150**, the endothermic coolant **200** flows to the thermally managed control module **122** via the coolant supply channel **132** to cool the thermally managed control module **122**.

(45) Accordingly, the temperature of the thermally managed control module **122** decreases, such as to less than the predetermined temperature. Some of the endothermic coolant **200** may be released

to the coolant release reservoir **136** via the coolant release channel **134**. For example, 200 milliliters or another suitable volume may be released to the coolant release reservoir **136**.
(46) FIG. **5** is a schematic diagram of an example cooling system **220**.
(47) The cooling system **220** includes a thermally managed control module **222**, a first cooling system **224**, and a second cooling system **226**.
(48) The thermally managed control module **222** may be the same as the thermally managed control modules **102**, **122**. The thermally managed control module **222** includes a cold plate **230**. A coolant channel **232** extends through the cold plate **230** and is configured to carry coolant to cool the cold plate **230**, and thereby cool the thermally managed control module **222**.
(49) The first cooling system **224** includes a surge tank **236**, an overflow tank **238**, and a pump **240**. The surge tank **236** is positioned outside the thermally managed control module **222** and in fluid communication with the coolant channel **232** via a coolant release channel **244**. The overflow tank **238** is in fluid communication with the surge tank **236** via an overflow channel **246** and receives overflow of coolant from the surge tank **236**. The pump **240** is positioned between a tank outlet **248** of the surge tank **236** and a channel inlet **250** of the coolant channel **232**. The pump **240** is in fluid communication with the surge tank **236** via a coolant return channel **252** and in fluid communication with the channel inlet **250** of the coolant channel **232** via a coolant supply channel **254**. A first check valve **258** is positioned between the pump **240** and the channel inlet **250** of the coolant channel **232**.
(50) Coolant is configured to flow from the coolant channel **232** to the surge tank **236** via the coolant release channel **244**. At least some coolant flows from the surge tank **236** to the pump **240** via the coolant return channel **252**. Some coolant may flow from the surge tank **236** to the overflow tank **238** via the overflow channel **246**. The pump **240** pumps coolant from the pump **240** to the channel inlet **250** of the coolant channel **232** via the coolant supply channel **254**. The first check valve **258** maintains a flow direction of coolant from the pump **240** to the channel inlet **250** of the coolant channel **232**. The coolant cools the cold plate **230** of the thermally managed control module **222**, and thereby cools the thermally managed control module **222**.
(51) The second cooling system **226** includes a fluid tank **264**, a gas cylinder **266**, and a release valve **268**. The fluid tank **264** contains a fluid. In one example, the fluid may be water. A solute reservoir **270** is disposed within the fluid tank **264** and contains a solute (e.g., endothermic solute). The solute may be ammonium nitrite, urea, or another suitable solute. An agitator **272** may be disposed within the fluid tank **264** and may be positioned adjacent to the solute reservoir **270**. The fluid tank **264** is in fluid communication with the coolant supply channel **254** via an endothermic coolant channel **276**. The endothermic coolant channel **276** may join with the coolant supply channel **254** in a position between the first check valve **258** and the cold plate **230**. A second check valve **278** may be positioned within the endothermic coolant channel **276**.
(52) The gas cylinder **266** is filled with a gas (e.g., pressurized gas). A gas channel **282** is positioned between the gas cylinder **266** and the solute reservoir **270**. The release valve **268** is disposed within the gas channel **282**. The release valve **268** is movable between an open position and a closed position. When the release valve **268** is in the open position, the gas cylinder **266** is in fluid communication with the solute reservoir **270** via the gas channel **282** and allows a flow of gas from the gas cylinder **266** to the solute reservoir **270**. When the release valve **268** is in the closed position, the release valve **268** restricts the flow of gas between the gas cylinder **266** and the solute reservoir **270**.
(53) When a temperature of the thermally managed control module **222** is below a predetermined temperature, coolant flows through the first cooling system **224**.
(54) When the temperature of the thermally managed control module **222** is greater than the predetermined temperature, an actuator is configured to transition the release valve **268** from the closed position to the open position based on the temperature. Gas from the gas cylinder **266** flows through the gas channel **282**, enters the fluid tank **264**, and ruptures the solute reservoir **270**. The

solute in the solute reservoir **270** is released from the solute reservoir **270** and mixes with the fluid from the fluid tank **264** using the agitator **272** to form an endothermic coolant via an endothermic reaction between the solute and the fluid. The endothermic coolant flows from the fluid tank **264** to the coolant channel **232** in the cold plate **230** via the endothermic coolant channel **276** and the coolant supply channel **254**. The endothermic coolant cools the cold plate **230** of the thermally managed control module **222**, and thereby cools the thermally managed control module **222**. Accordingly, the temperature of the thermally managed control module **222** returns to a temperature below the predetermined temperature. The endothermic coolant flows from the coolant channel **232** to the surge tank **236** via the coolant release channel **244** and some endothermic coolant may subsequently flow from the surge tank **236** to the overflow tank **238** via the overflow channel **246**.

(55) FIG. **6** is a schematic diagram of an example cooling system **320**.

(56) The cooling system **320** may be the same as the cooling system **220**, with the exception that the cooling system **320** includes a mechanical valve **316** and a second coolant channel **318**.

(57) The cooling system **320** includes a thermally managed control module **322**, a first cooling system **324**, and a second cooling system **326**. The thermally managed control module **322** may include a cold plate **330** and a coolant channel **332** extending through the cold plate **330**.

(58) The first cooling system **324** includes a surge tank **336**, and overflow tank **338**, and a pump **340**. The pump **340** pumps coolant from the surge tank **336** to a channel inlet **350** of the coolant channel **332** via the coolant supply channel **354**. A first check valve **358** is disposed between the pump **340** and the channel inlet **350** of the coolant channel **332**.

(59) The second cooling system **326** includes a fluid tank **364** containing fluid, a gas cylinder **366** containing gas, a release valve **368** movable between open and closed positions, a solute reservoir **370** containing a solute, and an agitator **372**. An endothermic coolant channel **376** carries endothermic coolant from the fluid tank **364** to the coolant supply channel **354**.

(60) The mechanical valve **316** is disposed within the coolant supply channel **354**. The mechanical valve **316** is positioned between the pump **340** and the first check valve **358**. The mechanical valve **316** is movable between an open valve position and a closed valve position. In some configurations, the mechanical valve **316** may be a wax valve. When a coolant temperature of the coolant is greater than a predetermined coolant temperature and the coolant passes the mechanical valve **316**, the wax is configured to melt and thereby moves the mechanical valve **316** from the closed position to the open position.

(61) When the mechanical valve **316** is in the closed position, coolant flows from the pump **340** to the channel inlet **350** of the coolant channel **332** via the coolant supply channel **354**. When the mechanical valve **316** is in the open position, coolant flows from the coolant supply channel **354** to the solute reservoir **370** via the second coolant channel **318**.

(62) When a temperature of the thermally managed control module **322** is below a predetermined temperature, the coolant temperature of the coolant is below the predetermined coolant temperature, and coolant thereby flows through the first cooling system **324**.

(63) When the temperature of the thermally managed control module **322** is greater than the predetermined temperature, the coolant temperature of the coolant is greater than the predetermined coolant temperature. Because the coolant temperature is greater than the predetermined coolant temperature, the coolant causes the mechanical valve **316** to transition from the closed position to the open position. Coolant flows from the mechanical valve **316** to the solute reservoir **370** via the second coolant channel **318**. The solute in the solute reservoir **370** is released from the solute reservoir **370** and mixes with the fluid from the fluid tank **364** using the agitator **372** to form an endothermic coolant via an endothermic reaction. The endothermic coolant flows from the fluid tank **364** to the coolant channel **332** in the cold plate **230** via the endothermic coolant channel **376** and the coolant supply channel **354**. Endothermic coolant cools the cold plate **330** of the thermally managed control module **322**, and thereby cools the thermally managed control module **322** such

that the temperature of the thermally managed control module **322** returns to a temperature below the predetermined temperature. The endothermic coolant flows from the coolant channel **332** to the surge tank **336** and some endothermic coolant may subsequently flow from the surge tank **336** to the overflow tank **338**.
(64) FIG. **7** is a schematic diagram of an example vehicle **400**. The vehicle includes an airbag **402**, an airbag control module **404**, and a cooling system **420**.
(65) The cooling system **420** may be the same as the cooling system **220**, with the exception that the cooling system **420** includes a bypass trigger **416** and a second gas channel **418**.
(66) The cooling system **420** includes a thermally managed control module **422**, a first cooling system **424**, and a second cooling system **426**. The thermally managed control module **422** may include a cold plate **430** and a coolant channel **432** extending through the cold plate **430**.
(67) The first cooling system **424** includes a surge tank **436**, an overflow tank **438**, and a pump **440**. The pump **440** pumps coolant from the surge tank **436** to a channel inlet **450** of the coolant channel **432** via a coolant supply channel **454**. A first check valve **458** is disposed between the pump **440** and the channel inlet **450** of the coolant channel **432**.
(68) The second cooling system **426** includes a fluid tank **464** containing fluid, a gas cylinder **466** containing gas, and a release valve **468** movable between open and closed positions. The fluid tank **464** includes a solute reservoir **470** containing a solute and an agitator **472**. An endothermic coolant channel **476** is configured to carry endothermic coolant from the fluid tank **464** to the coolant supply channel **454**. A gas channel **482** extends between the gas cylinder **466** and the solute reservoir **470**.
(69) The bypass trigger **416** is positioned parallel with the release valve **468**. More specifically, the bypass trigger **416** is disposed within the second gas channel **418** to bypass the release valve **468**. The second gas channel **418** extends between a first channel end **484** and a second channel end **486** that opposes the first channel end **484**. The second gas channel **418** is in fluid communication with the gas channel **482**. The first channel end **480** is positioned between the gas cylinder **466** and the release valve **468**. The second channel end **486** is positioned between the release valve **468** and the fluid tank **464**.
(70) When the airbag **402** of the vehicle **400** is not activated, coolant flows through the first cooling system **424**.
(71) When the airbag **402** of the vehicle **400** is activated, the airbag control module **404** is configured to activate the bypass trigger **416** based on the activated state of the airbag **402**. The bypass trigger **416** opens the second gas channel **418** and allows gas to flow from the gas cylinder **466** to the solute reservoir **470** via the second gas channel **418**. The solute is released from the solute reservoir **470** and mixes with the fluid from the fluid tank **464** using the agitator **472** to form an endothermic coolant via an endothermic reaction. The endothermic coolant flows from the fluid tank **464** to the coolant channel **432** in the cold plate **430** via the endothermic coolant channel **476** and the coolant supply channel **454**. Endothermic coolant cools the cold plate **430** of the thermally managed control module **422**, and thereby cools the thermally managed control module **422**. The endothermic coolant flows from the coolant channel **432** to the surge tank **436** and some endothermic coolant may subsequently flow from the surge tank **436** to the overflow tank **438**.
(72) FIG. **8** is a schematic diagram of an example cooling system **520**.
(73) The cooling system **520** may be the same as the cooling system **220**, with the exception that the cooling system **520** includes a second cold plate **516** with a second coolant channel **518**.
(74) The cooling system **520** includes a thermally managed control module **522**, a first cooling system **524**, and a second cooling system **526**. The thermally managed control module **522** includes a cold plate **530** and a coolant channel **532** extending through the cold plate **530**. The thermally managed control module **522** includes the second cold plate **516** and the second coolant channel **518** extending through the second cold plate **516**. The cold plate **530** and the second cold plate **516** are disposed at opposing ends of the thermally managed control module **522**.

(75) The first cooling system **524** includes a surge tank **536** and a pump **440**. The pump **440** pumps coolant from the surge tank **436** to the coolant channel **432** via a coolant supply channel **554**.
(76) The second cooling system **526** includes a fluid tank **564** containing fluid, a gas cylinder **566** containing gas, and a release valve **568** movable between open and closed positions. The fluid tank **564** includes a solute reservoir **570** containing a solute and an agitator **572**. An endothermic coolant supply channel **576** is configured to carry endothermic coolant from the fluid tank **564** to the second coolant channel **518**. A gas channel **582** extends between the gas cylinder **566** and the solute reservoir **570**.
(77) The endothermic coolant supply channel **576** is configured to carry endothermic coolant from the fluid tank **564** to cool the thermally managed control module **522**. A second coolant release channel **586** extends between the second coolant channel **518** and an overflow tank **538**.
(78) When a temperature of the thermally managed control module **522** is below a predetermined temperature, coolant flows from the coolant channel **532** to the surge tank **536**. At least some coolant flows from the surge tank **536** to the pump **540**. The pump **540** pumps coolant from the pump **540** to a channel inlet **550** of the coolant channel **532** via the coolant supply channel **554**. The coolant cools the cold plate **530** of the thermally managed control module **522**, and thereby cools the thermally managed control module **522**.
(79) When the temperature of the thermally managed control module **522** is greater than the predetermined temperature, an actuator is configured to transition the release valve **568** from the closed position to the open position based on the temperature. Gas from the gas cylinder **566** flows through the gas channel **582**, enters the fluid tank **564**, and ruptures the solute reservoir **570**. The solute in the solute reservoir **570** is released from the solute reservoir **570** and mixes with the fluid from the fluid tank **564** using the agitator **572** to form the endothermic coolant via an endothermic reaction between the solute and the fluid. The endothermic coolant flows from the fluid tank **564** to the second coolant channel **518** in the second cold plate **516** via the endothermic coolant supply channel **576**. The endothermic coolant cools the second cold plate **516** of the thermally managed control module **522**, and thereby cools the thermally managed control module **522**. The endothermic coolant flows from the second coolant channel **518** to an overflow tank **538** via the second coolant release channel **586**. Accordingly, the temperature of the thermally managed control module **522** returns to a temperature below the predetermined temperature.
(80) FIG. **9** is a schematic diagram of an example cooling system **620**.
(81) The cooling system **620** may be the same as the cooling system **220**, with the exception that the cooling system **620** does not include a fluid tank.
(82) The cooling system **620** includes a thermally managed control module **622**, a first cooling system **624**, and a second cooling system **626**.
(83) The thermally managed control module **622** includes a cold plate **630** and a coolant channel **632** extending through the cold plate **630**. The first cooling system **624** includes a coolant supply channel **654**. The second cooling system **626** includes a gas cylinder **666**, a release valve **668**, a solute reservoir **670**, and an agitator **672**. The fluid tank **664** is in fluid communication with the coolant supply channel **654** via an endothermic coolant channel **676**.
(84) When the temperature of the thermally managed control module **622** is less than the predetermined temperature, coolant cools the thermally managed control module **622** using the first cooling system **624**.
(85) When the temperature of the thermally managed control module **622** is greater than the predetermined temperature, an actuator moves the release valve **668** from a closed position to an open position. Gas travels from the gas cylinder **666** to the solute reservoir **670** and ruptures the solute reservoir **670**. The solute in the solute reservoir **670** is released from the solute reservoir **670** and travels to the endothermic coolant channel **676** and the coolant supply channel **654**. The solute mixes with coolant fluid within the coolant supply channel **654** to create an endothermic coolant via an endothermic reaction. The endothermic coolant flows from coolant supply channel **654** to

the coolant channel **632** in the cold plate **630**. The endothermic coolant cools the cold plate **630** of the thermally managed control module **622**, and thereby cools the thermally managed control module **622** such that the temperature of the thermally managed control module **622** returns to a temperature below the predetermined temperature.

(86) FIG. **10** is a schematic diagram of an example cooling system **720**.

(87) The cooling system **720** includes a thermally managed control module **722**, a first cooling system **724**, and a second cooling system **726**. The thermally managed control module **722** and the second cooling system **726** are disposed within a fluid tank **728**. The fluid tank **728** contains a fluid. In one example, the fluid may be water.

(88) The first cooling system **724** may be the same as the first cooling system **224**. A coolant channel **732** is disposed within the fluid tank **728** and is in fluid communication with the first cooling system **724**.

(89) The second cooling system **726** includes a solute reservoir **770** and a release assembly **772**. The solute reservoir **770** is disposed within the fluid tank **728** and contains a solute. The solute may be ammonium nitrite, urea, or another suitable solute. The release assembly **772** may be disposed within the fluid tank **728** and may be positioned adjacent to the solute reservoir **770**. The release assembly **772** will be described in greater detail below and with reference to FIGS. **12-14**.

(90) When a temperature of the thermally managed control module **722** is below a predetermined temperature, coolant flows through the coolant channel **732** using the first cooling system **724**. The coolant flowing through the coolant channel **732** extracts heat from the fluid of the fluid tank **728** and thereby, reduces a fluid temperature of the fluid. The fluid cools the thermally managed control module **722** such that the temperature of the thermally managed control module **722** remains below the predetermined temperature. In one example, the fluid cools the thermally managed control module **822** by convection. In another example, the fluid cools the thermally managed control module **822** by forced advection.

(91) When the temperature of the thermally managed control module **722** is greater than the predetermined temperature, the thermally managed control module **722** activates the release assembly **772**. The release assembly **772** releases the solute from within the solute reservoir **770** into the fluid of the fluid tank. The solute mixes with the fluid to create an endothermic coolant via an endothermic reaction. The endothermic coolant cools the thermally managed control module **722** and returns the temperature of the thermally managed control module **722** to below the predetermined temperature.

(92) FIG. **11** is a schematic diagram of an example cooling system **820**.

(93) The cooling system **820** includes a thermally managed control module **822**, a first cooling system **824**, and a second cooling system **826**. The thermally managed control module **822** is disposed within a fluid tank **828**.

(94) The fluid tank **828** extends between a top end **836** and a bottom end **838**. The fluid tank **828** is partially filled with a fluid from the bottom end **838**. In one example, the fluid may be ethanol, a refrigerant, or another suitable fluid. Gas is disposed within the fluid tank **828** and positioned at the top end **836** of the fluid tank **828**. The thermally managed control module **822** is disposed within the fluid of the fluid tank **828**.

(95) The first cooling system **824** may be the same as the first cooling system **224**. A coolant channel **832** is disposed within the fluid of the fluid tank **828** and is in fluid communication with the first cooling system **824**.

(96) The second cooling system **826** includes a gas channel **842**, a vent **844**, and pressure relief valve **846**. The vent **844** is positioned outside the fluid tank **828**. The vent **844** is in communication with the top end **836** of the fluid tank **828** via the gas channel **842**.

(97) The pressure relief valve **846** is disposed within the gas channel **842** to restrict and controls flow of gas from the fluid tank **828** to the vent **844**. The pressure relief valve **846** is movable between an open position and a closed position. When a pressure in the gas channel **842** between

the fluid tank **828** and the pressure relief valve **846** is below a predetermined pressure (e.g., a cracking pressure), the pressure relief valve **846** is in the closed position and restricts flow of gas from the fluid tank **828** to the vent **844**. When the pressure in the gas channel **842** between the fluid tank **828** and the pressure relief valve **846** is above the predetermined pressure, the pressure relief valve **846** opens to the open position and allows release of gas from the fluid tank **828** and out the vent **844**.

(98) When a temperature of the thermally managed control module **822** is below a predetermined temperature, coolant flows through the coolant channel **832** using the first cooling system **824**. The coolant flowing through the coolant channel **832** extracts heat from the fluid of the fluid tank **828** and thereby, reduces a fluid temperature of the fluid. The fluid cools the thermally managed control module **822** such that the temperature of the thermally managed control module **822** remains below the predetermined temperature. In one example, the fluid cools the thermally managed control module **822** by convection. In another example, the fluid cools the thermally managed control module **822** by forced advection.

(99) When the temperature of the thermally managed control module **822** reaches the predetermined temperature, the fluid temperature of the fluid reaches a saturation point of the fluid and begins to evaporate. In other words, the fluid is converted into a gas. The conversion from fluid into a gas increases the pressure in the fluid tank **828** and in the gas channel **842** between the fluid tank **828** and the pressure relief valve **846**. More specifically, the pressure is greater than the predetermined pressure and the pressure relief valve **846** moves from the closed position to the open position. Subsequently, gas flows from the fluid tank **828**, through the gas channel **842**, and out the vent **844**. Accordingly, release of gas from the fluid tank **828** reduces the fluid temperature of the fluid in the fluid tank **828** and reduces the temperature of the thermally managed control module **822** such that the temperature of the thermally managed control module **822** is at or below the predetermined temperature.

(100) FIG. **12** is a schematic diagram of an example release system **900** for an example solute reservoir **902**. The release system **900** may replace the agitator **272**, **372**, **472**, **672** of the cooling system **220**, **320**, **420**, **620**, respectively, and may be the release assembly **772** of the cooling system **720**.

(101) The solute reservoir **902** is disposed in a fluid tank **904**. The fluid tank **904** is filled with a fluid and the solute reservoir **902** contains a solute. The solute reservoir **902** includes sidewalls **906** that are made of a soft membrane.

(102) The release system **900** includes an impeller **908**, a motor **910**, and an arm **912**. The impeller **908** is in contact with one of the sidewalls **906** of the solute reservoir **902** and is rotatably relative to the solute reservoir **902**. The motor **910** is electrically connected to the impeller **908** and powers rotational movement of the impeller **908**. The arm **912** is fixed to the motor **910** and positions the impeller **908** adjacent to the one of the sidewalls **906**.

(103) Activation of the motor **910** drives rotational movement of the impeller **908** and rotational movement of the impeller **908** pierces the one of the sidewalls **906** of the solute reservoir **902** to release the solute into the fluid tank **904**. Further rotational movement of the impeller **908** functions as an agitator and mixes the solute with the fluid. The solute and the fluid mix under an endothermic reaction and creates an endothermic coolant via an endothermic reaction.

(104) FIG. **13** is a schematic diagram of an example release system **930** for an example solute reservoir **932**. The release system **930** may replace the agitator **272**, **372**, **472**, **672** of the cooling system **220**, **320**, **420**, **620**, respectively, and may be the release assembly **772** of the cooling system **720**.

(105) The solute reservoir **932** is disposed in a fluid tank **934**. The fluid tank **934** is filled with a fluid and the solute reservoir **932** contains a solute. The solute reservoir **932** includes sidewalls **936** that are made of a soft membrane.

(106) The release system **930** includes a magnet **940**, a solenoid **942**, and an arm **944**. The magnet

940 is disposed within the solute reservoir **932**. The solenoid **942** is disposed outside the solute reservoir **932** and within the fluid tank **934**. The solenoid **942** is positioned adjacent to the magnet **940**. More specifically, the solenoid **942** may be aligned with the magnet **940**. The arm **944** is fixed to the solenoid **942** and holds the solenoid **942** adjacent to the magnet **940**.

(107) Activation of the solenoid **942** drives movement of the magnet **940**. More specifically, the solenoid **942** attracts the magnet **940** and the magnet **940** pierces through the one of the sidewalls **936** of the solute reservoir **932** to release the solute into the fluid tank **934**. The solute and the fluid mix to create an endothermic coolant via an endothermic reaction.

(108) FIG. **14** is a schematic diagram of an example release system **960** for an example solute reservoir **962**. The release system **960** may replace the agitator **272**, **372**, **472**, **672** of the cooling system **220**, **320**, **420**, **620**, respectively, and may be the release assembly **772** of the cooling system **720**.

(109) The solute reservoir **962** is disposed in a fluid tank **964**. The fluid tank **964** is filled with a fluid and the solute reservoir **962** contains a solute. The solute reservoir **962** includes sidewalls **966** that are made of a soft membrane.

(110) The release system **960** includes a fuse **970** that extends through at least one sidewall **966** of the solute reservoir **962**. In the illustrated example, the fuse **970** extends through two sidewalls **966** that are positioned at opposite ends of the solute reservoir **962**.

(111) Current travels through the fuse **970** and is operable to melt the at least one sidewall **966** of the solute reservoir **962** to release the solute into the fluid tank **964**. The solute and the fluid mix to create an endothermic coolant via an endothermic reaction.

(112) FIG. **15** is a flow chart of an example method **1000** for operating the cooling system. The cooling system may be any of the cooling systems **120**, **220**, **320**, **420**, **520**, **620**, **720**, **820**.

(113) At **1004**, the thermally managed control module determines whether a temperature of the thermally managed control module is greater than a predetermined temperature. If **1004** is not true (e.g., the temperature does not exceed the predetermined temperature), the method continues to **1008**. If **1004** is true (e.g., the temperature is greater than the predetermined temperature), the method continues to **1012**.

(114) At **1008**, the first cooling system cools the thermally managed control module.

(115) At **1012**, the second cooling system begins to cool the thermally managed control module. In some examples, the second cooling system may cool the thermally managed control module for a time period. The time period may be a time period of about 5 minutes, or another suitable time period.

(116) At **1016**, the thermally managed control module notifies a host control module of the vehicle based on the temperature of the thermally managed control module.

(117) At **1020**, the host control module brings the vehicle to a predetermined state based on the notification received from the thermally managed control module. For example, the host control module steers the vehicle to a location, such as a nearby shoulder of a road, a nearby parking lot, or another suitable location.

(118) At **1024**, the second cooling system is replaced by an owner of the vehicle. In some examples, the pressurized gas cylinder is replaced. In some examples, the second cooling system are flushed of the endothermic coolant with a second fluid. In some examples, the fluid of the fluid tank is replaced with a third fluid.

(119) The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described

above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(120) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including "connected," "engaged," "coupled," "adjacent," "next to," "on top of," "above," "below," and "disposed." Unless explicitly described as being "direct," when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean "at least one of A, at least one of B, and at least one of C."

(121) In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

(122) In this application, including the definitions below, the term "module" or the term "controller" may be replaced with the term "circuit." The term "module" may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(123) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

(124) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, data structures, and/or objects. The term shared processor circuit encompasses a single processor circuit that executes some or all code from multiple modules. The term group processor circuit encompasses a processor circuit that, in combination with additional processor circuits, executes some or all code from one or more modules. References to multiple processor circuits encompass multiple processor circuits on discrete dies, multiple processor circuits on a single die, multiple cores of a single processor circuit, multiple threads of a single processor circuit, or a combination of the above. The term shared memory circuit encompasses a single memory circuit that stores some or all code from multiple modules. The term group memory circuit encompasses a memory circuit that, in combination with additional memories, stores some or all code from one or more modules.

(125) The term memory circuit is a subset of the term computer-readable medium. The term

computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).
(126) The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.
(127) The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.
(128) The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C#, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5.sup.th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

Claims

1. A cooling system comprising: a reactant material; a reactive material spaced apart from the reactant material; a thermally managed control module configured to receive coolant from a coolant supply reservoir via a coolant supply channel; a plunger movable between a retracted position and an extended position; and an actuator configured to, when a temperature of the thermally managed control module is greater than a predetermined temperature, transition the plunger from the retracted position to the extended position and cause the reactive material to mix with the reactant material to create a second coolant via an endothermic reaction, and the second coolant enters the thermally managed control module via the coolant supply channel.
2. The cooling system of claim 1, further comprising a plunger channel in fluid communication with the coolant supply channel and housing the plunger and the actuator.
3. The cooling system of claim 2, wherein: the actuator is a ram; and when the temperature of the thermally managed control module is greater than the predetermined temperature, the ram drives movement of the plunger from the retracted position to the extended position.
4. The cooling system of claim 2, further comprising: a first sealing disk disposed between the reactant material and the reactive material; and a second sealing disk disposed between the reactive material and the coolant supply channel.
5. The cooling system of claim 4, wherein: the plunger is disposed within the plunger channel when

in the retracted position, and the plunger extends transversely across the coolant supply channel and ruptures the first and second sealing disks when transitioning from the retracted position to the extended position.

6. The cooling system of claim 5, wherein the plunger includes a channel and the channel carries the second coolant to the coolant supply channel when the plunger is in the extended position.

7. A cooling system comprising: a fluid tank containing a fluid; a solute reservoir disposed within the fluid tank and containing an endothermic solute; a thermally managed control module having a coolant channel extending through the thermally managed control module and the coolant channel configured to receive a coolant; and an actuator configured to, when a temperature of the thermally managed control module is greater than a predetermined temperature, release the endothermic solute from the solute reservoir into the fluid tank and cause the endothermic solute to mix with the fluid to create a second coolant via an endothermic reaction, and the second coolant enters the thermally managed control module via a coolant supply channel.

8. The cooling system of claim 7, further comprising: a gas cylinder attached to the solute reservoir and filled with a pressurized gas; and a release valve disposed between the gas cylinder and the solute reservoir and movable between an open position and a closed position, wherein when the temperature of the thermally managed control module is greater than the predetermined temperature, the actuator is configured to transition the release valve from the closed position to the open position and allow the pressurized gas to flow from the gas cylinder to the solute reservoir and release the endothermic solute from the solute reservoir into the fluid.

9. The cooling system of claim 8, further comprising: a bypass trigger disposed between the gas cylinder and the solute reservoir, an airbag control module configured to activate the bypass trigger based on an activated state of an airbag, wherein when the bypass trigger is activated, the bypass trigger is configured to allow the pressurized gas to flow from the gas cylinder to the solute reservoir and release the endothermic solute from the solute reservoir into the fluid tank to create the second coolant via the endothermic reaction, and the second coolant enters the coolant channel of the thermally managed control module via the coolant supply channel.

10. The cooling system of claim 7, further comprising: a pump disposed outside the thermally managed control module and configured to pump the coolant from an outlet of the coolant channel to an inlet of the coolant channel.

11. The cooling system of claim 10, further comprising: a mechanical valve disposed between the pump and the inlet of the coolant channel and movable between an open position and a closed position, wherein when a coolant temperature of the coolant is above a predetermined coolant temperature, the coolant transitions the mechanical valve from the closed position to the open position and causes the endothermic solute to release from the solute reservoir into the fluid tank to create the second coolant via the endothermic reaction, and the second coolant enters the coolant channel of the thermally managed control module via the coolant supply channel.

12. The cooling system of claim 7, wherein the thermally managed control module includes a cold plate and the coolant channel extends through the cold plate.

13. The cooling system of claim 12, wherein the thermally managed control module includes a second coolant channel extending through a second plate and the second coolant channel is configured to receive the second coolant from the fluid tank via the coolant supply channel.

14. A cooling system comprising: a fluid tank at least partially filled with a fluid; the fluid has a fluid temperature; a thermally managed control module disposed within the fluid tank; a coolant channel extending through the fluid tank and configured to receive a coolant to cool the fluid in the fluid tank; and an actuator configured to, when the fluid temperature of the fluid is greater than a predetermined temperature, cool the fluid temperature to less than the predetermined temperature by at least one of evaporating the fluid and mixing the fluid with an endothermic solute.

15. The cooling system of claim 14, further comprising: a solute reservoir disposed within the fluid tank and containing the endothermic solute, wherein when the fluid temperature is greater than the

predetermined temperature, the actuator is configured to release the endothermic solute from the endothermic solute into the fluid of the fluid tank using a release assembly.
16. The cooling system of claim 15, wherein the release assembly includes an impeller disposed adjacent to the solute reservoir and rotation of the impeller pierces a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
17. The cooling system of claim 15, wherein: the release assembly includes a magnet disposed inside the solute reservoir and a solenoid disposed outside the solute reservoir, and the solenoid attracts the magnet and pierces a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
18. The cooling system of claim 15, wherein the release assembly includes a fuse disposed inside the solute reservoir, and current traveling through the fuse is configured to melt a sidewall of the solute reservoir such that the endothermic solute is released into the fluid of the fluid tank.
19. The cooling system of claim 14, wherein the fluid tank is at least partially filled with a gas.
20. The cooling system of claim 19, further comprising: a pressure relief valve movable between an open position and a closed position, wherein when the fluid temperature is greater than the predetermined temperature, the fluid in the fluid tank evaporates and the gas causes the pressure relief valve to transition from the closed position into the open position to release the gas from the fluid tank via a vent.